

	Kinetic energies at Q and R are the same.
12.	Ans: B Intensity = Power / Area Intensity at 4.0 m from source, $I = 50.0 / [4\pi(4.0)^2] = 0.2487 \text{ W m}^{-2}$ Power detected by microphone = $I \times \text{Mic Area} = (0.2487)(0.85 \times 10^{-4}) = 2.11 \times 10^{-5} \text{ W}$
13.	Ans: B